DESIGN CARD · HACK THE HIGHWAY · TAMU DATATHON 2026

# Wacky Wheels Control Tower

One coordinator + three specialist agents turn a messy operations feed into policy-checked actions. Monday 8 March 2027 · one fresh world per event.

| 15 | 909 | 0 | 12/15 | 3B |
|---|---|---|---|---|
| EVENTS | TOOL CALLS | FAILED | LLM AGREED | LLAMA 3.2 |

## 1 ARCHITECTURE: SIX GATES PER MESSAGE

**01 — The wire**
One message. It may hold two incidents, a rumor, or nothing at all.

**02 · LLM — Llama reads**
Proposes the incident kind, the names and the date phrases as one JSON object.

**03 — The graph binds**
Aliases, the working calendar, and exactly one shipment or lot.

**04 — Coordinator**
Dispatches the desks, stands down for noise, or escalates.

**05 — Specialists**
Walk the ontology and return a *plan*, never a tool call.

**06 · TOOLS — The gateway**
Every argument is checked, then the enterprise tools run.

Plant outage · Supplier delay · Quality hold → full dispatch

Noise → stand down, no call

Unconfirmed → `escalate` only

Compound → one shared slot board + loaner pool per event

## 2 AGENTS AND THEIR TOOLS

**Coordinator**
Splits the message into incidents. The only component that sends tool calls.
`escalate`

**Supply Chain Planner**
Earliest open slot. Transfers only when the home plant is full; ties follow handbook order.
`mes.reschedule` `mes.transfer`

**Quality Sentinel**
Holds the lot, stops sale of unsold vehicles, contacts owners of sold ones.
`quality.hold_lot`
`dealer.stop_sale`
`crm.notify_customer`

**Dealer & Customer Desk**
One notice per dealer. Loaners: Platinum → Gold → earlier promise → lower order id.
`dealer.notify`
`crm.loaner_availability`
`crm.assign_loaner`

Specialists return dictionaries. `Coordinator.handle` assembles the action list; only the gateway in `run.py` calls the server.

## 3 LANGUAGE MODEL VS. CODE

**Llama 3.2 3B proposes**
- Incident kind & disposition
- Plant, part, supplier names
- Ids only if written in the text
- Raw date phrases

**Code decides**
- Policy rules & calendar
- Capacity, slots, transfers
- Loaner priority & caps
- **Every** tool argument

01 02 03 04 05 06 07 08 09 10 11 12 13 14 15

agreed 12 events · ontology reader kept E-01, E-03, E-09.

Tool arguments are identical either way: the model never chooses an order, a VIN or a slot.

## 4 GUARDRAILS

| | |
|---|---|
| Noise | No call at all. The trace cites the rule and why the text is out of scope. |
| Rumor | `escalate` only. No order, dealer, loaner, lot or customer is touched. |
| Check → send | Plan rehearsed on a private `Enterprise` first. A rejected call is logged, not retried. |
| One board | Compound events share slots and loaners, so nothing is booked twice. |
| Precision | One `dealer.notify` per dealer listing only its orders. Standard tier gets no loaner. Loaner response caps assignments. |

## 5 ONTOLOGY PATHS CITED PER INCIDENT

**Supplier delay** `Shipment → Part → Model (BOM) → ProductionOrder → CustomerOrder → Dealer / Customer`
**Plant outage** `Plant → ProductionOrder → CustomerOrder → Dealer / Customer`
**Quality hold** `Lot → ProductionOrder → Vehicle → Dealer / Customer`

13 entity types · ~17k nodes (540 production orders, 3,263 customer orders, 7,000 customers, 5,348 vehicles).

## 6 EVALUATION

`run.py` replays the full plan on a private `Enterprise` before the server starts, checking locations, capacity, notices, loaners, held lots, stopped VINs, owner contacts, escalation, and that unaffected orders did not move.

Scored file: `runs/run1/action_log.jsonl` (909 lines, 0 failures).

## 7 WHAT WE WOULD IMPROVE

- A larger local model to close E-01, E-03 and E-09, still never allowed to pick a slot.
- A published plant tie-break in the handbook so every team lands the same transfer.

**Model:** Llama 3.2 3B (`llama3.2:3b`, Meta) via Ollama on a Mac, no API key, no cloud inference.

github.com/srijapentyala/hack_the_highway